

Tier 3: Reaction-level reasoning

	LLM 100	CodeAct 100	RLM 100	LLM 500	CodeAct 500	RLM 500	RLM Full
Bond-level (3Q)	0.44	1.00	1.00	0.04	0.21	1.00	1.00
Scaffold (3Q)	0.57	0.56	0.77	0.05	0.08	0.71	0.75
Reagent (2Q)	0.75	0.87	0.94	0.22	0.21	0.68	0.87
Stereochemistry (2Q)	0.89	0.91	1.00	0.34	0.10	1.00	0.99
Functional group (15Q)	0.73	0.75	0.79	0.28	0.26	0.76	0.62
Mechanism (10Q)	0.72	0.64	0.70	0.50	0.27	0.66	0.39

Tier 4: Relational reasoning

	LLM 100	CodeAct 100	RLM 100	LLM 500	CodeAct 500	RLM 500	RLM Full
Mechanical graph (5Q)	0.49	0.68	1.00	0.06	0.02	1.00	1.00
Chemical constraints (10Q)	0.08	0.60	0.92	0.00	0.08	0.86	0.79
Prospective / multi-constraint (20Q)	0.34	0.35	0.57	0.05	0.01	0.39	0.05

Macro F1